QAOA-Tuner: Noise- and Hardware-Aware Configuration Optimization for Quantum Graph Coloring

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Qiskit | QAOA | Graph Coloring | Noise-Aware Simulation

Scan for the code

Abstract

Quantum Approximate Optimization Algorithm (QAOA) is evaluated for graph coloring under hardware-aware transpilation and simulated noise. The study analyzes solution quality, circuit complexity, and error mitigation using Qiskit Aer, to identify effective configurations through automated evaluation and quality–cost trade-off analysis.

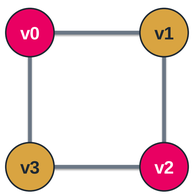
Problem & Motivation

QAOA performance depends on algorithmic depth, parameter optimization, circuit compilation, hardware connectivity, and noise. Different configurations can therefore produce different trade-offs between solution quality and execution cost. The challenge is to systematically evaluate these configurations rather than manually selecting one.

Objective

Build a Qiskit-based framework that automatically evaluates QAOA configurations and exposes quality–hardware-cost trade-offs through Pareto analysis.

Graph Coloring

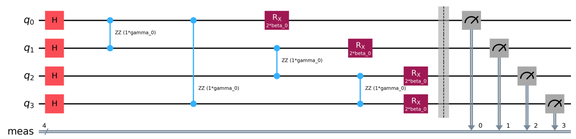


Valid 2-coloring of C4

- Assign colors to vertices so that adjacent vertices receive different colors.
- C4 (a 4-cycle) is the representative optimization problem.
- Example: 4 vertices, 2 colors.

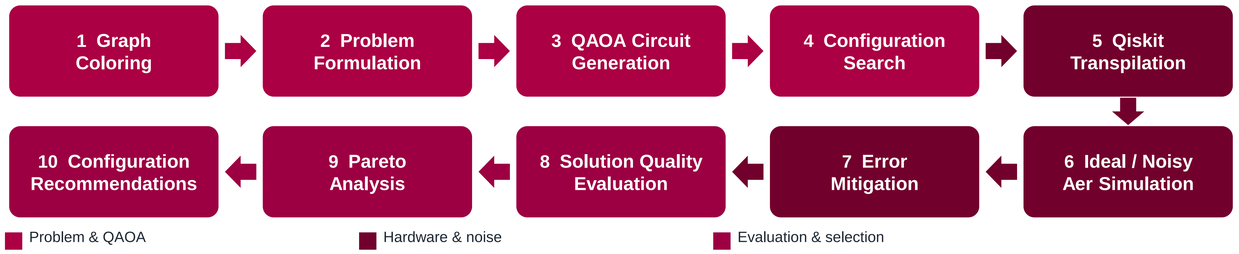
QAOA

- QAOA searches for low-cost graph-coloring assignments.
- Depth p sets the number of layers and the circuit complexity.
- Classical optimizers (COBYLA, SPSA) tune the parameters.
- Qiskit builds and transpiles the circuits.

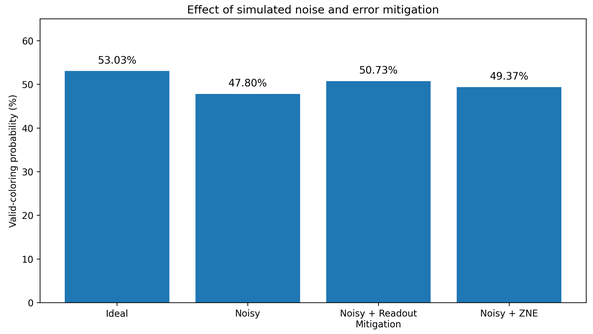


QAOA circuit for C4, p = 1: H, ZZ cost layer, RX mixer, measurement

Methodology: QAOA-Tuner Workflow



Main Results



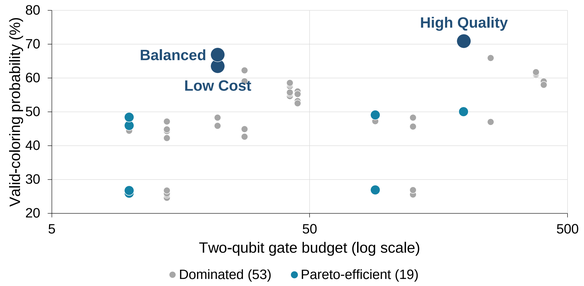
Ideal	Noisy	+ Readout	+ ZNE
53.03%	47.80%	50.73%	49.37%

Simulated noise reduces the valid-coloring probability by 5.23 % points; readout mitigation and ZNE recover part of this loss.

Single run: p = 1, COBYLA, 2048 shots, seed 42; Qiskit Aer simulated noise.

QAOA-Tuner: Pareto Analysis

72	19	10
configurations evaluated	Pareto-efficient	distinct trade-offs

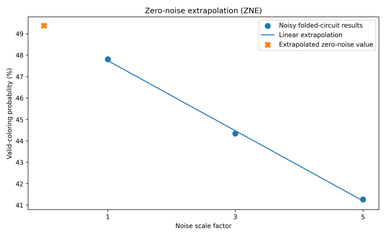


Low Cost	Balanced	High Quality
63.5%	66.9%	70.9%
2q gate budget 22	2q gate budget 22	2q gate budget 198

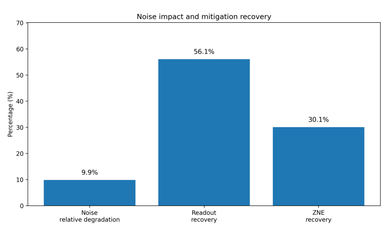
Descriptive trade-offs, not a single "best". All three: p = 2, COBYLA, transpiler level 1 (none / readout mitigation / ZNE).

Grid search: 1024 shots, seed 42; Qiskit Aer simulated noise.

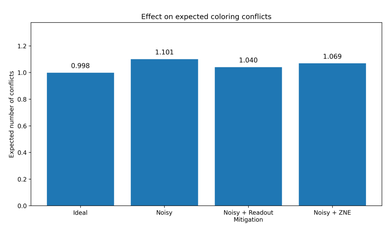
Error Mitigation: Readout and Zero-Noise Extrapolation (ZNE)



ZNE: noise scale factors 1, 3, 5; linear extrapolation



Noise degradation and share of the gap recovered



Expected coloring conflicts (lower is better)

ZNE zero-noise estimate 49.37% (±1.33%); relative noise-induced degradation 9.85%; gap recovered: readout 56.1%, ZNE 30.1%.

Single run (p = 1): the ZNE gain (+1.6 points) is comparable to its standard error (1.33 points), and readout mitigation has no standard-error estimate. Both techniques add overhead (ZNE: 9× the two-qubit gates) and neither removes all noise.

Hardware-Aware Evaluation

- Circuits are transpiled with Qiskit for the simulated backend fake_linear_5q.
- Measured per circuit: depth, 1- and 2-qubit gate counts, two-qubit gate budget, total shots.
- Noise is simulated using Qiskit Aer noise models; no physical quantum hardware is used.

p = 2: 8 logical ZZ gates → 22 two-qubit gates after transpilation.
Transpiler level 0 → 1: two-qubit gates 28 → 22, depth 51 → 40.

Key Findings

- Configurations show measurable quality–circuit-cost trade-offs under simulated noise.
- Higher transpiler optimization level reduced two-qubit gates.
- Readout mitigation and ZNE partially recover noise-induced degradation.
- Pareto analysis exposes configurations for different quality/cost priorities.
- Across all 6 seeds, ZNE achieved the highest valid-coloring rate, followed by readout mitigation and no mitigation (p = 2, COBYLA, transpiler level 1).
- QAOA-Tuner is a systematic Qiskit-based workflow for evaluating these trade-offs.

Limitations & Future Work

Limitations

- Qiskit Aer simulated noise models only
- No physical quantum hardware used
- Small graph instances (C4)
- Results vary with optimizer initialization and shot noise
- Larger multi-graph and multi-seed evaluations are future work

Future work

Multi-graph and larger multi-seed evaluation · Noise-aware parameter training · Dynamical decoupling · Per-qubit calibrated noise · Larger graph instances · Physical hardware validation

References

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Tech stack: Python | Qiskit | Qiskit Aer | NetworkX | NumPy | SciPy | Matplotlib | Streamlit

github.com/emjayy13/QAOA-Tuner